CohortCert: Shared Evidence and Proof Costs for Cohort Decisions

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***Abstract*—Many data analysis tasks involve comparing groups of records based on properties that require costly labeling by large language models (LLMs) or human annotators. These tasks often involve multiple questions about how frequently a property occurs in different groups. When groups overlap and comparisons share evidence, the challenge is to decide which records to label so that the entire request can be answered within budget. Labeling records separately for each comparison wastes shared evidence, while answering only some questions leaves the request incomplete. We present CohortCert, a framework that shares labeling evidence across groups and comparisons to answer entire requests efficiently. CohortCert organizes overlapping records by their contributions to the requested comparisons, maintains confidence intervals over their label counts, and adaptively selects records to label based on their contribution to completing the full request. We further develop an offline completion-cost analysis that uses verified upper and lower bounds to distinguish missed allocation opportunities from budget limitations. Across 73,760 saved-label executions, CohortCert improves completion rates over width-based allocation by 0.78 and 1.16 percentage points on CivilComments with crowd and LLM labels, respectively, while showing similar performance on GoEmotions. Offline analysis constructs within-budget proofs in 157 of 192 selected cases and establishes an evidence-cost advantage from jointly handling overlapping records in 30 cases.**

***Index Terms*—semantic query processing, finite-population inference, Boolean certificates, evidence acquisition, cohort comparison**

I. Introduction

An analyst studying online discussions may want to know whether toxic language is more common in one group of comments than another. A *group*, or cohort, is a set of records selected by known metadata: for example, comments that mention a particular topic. A *label* records whether one comment has the property of interest. The group's rate is the fraction of its records labelled positive. These definitions also describe other tasks, such as comparing emotion rates across discussion forums.

A request can contain several questions: which group has the higher rate, whether two rates differ by less than five percentage points, and which groups form the top two. The system obtains labels by reading existing annotations or invoking a fixed model. In the latter case, each unread label can require an expensive model call. A limited call budget makes the choice of records important. The output must include an answer to every requested question, supported by the evidence collected so far. When the evidence is insufficient, the system should say that the request remains incomplete.

Two kinds of sharing make this problem different from estimating each rate separately. A comment can belong to several groups, so its label should be acquired once and used in all relevant rates. Several questions can also depend on the same rate difference. The next read should therefore be chosen for its value to the *whole request*. Resolving one easy comparison does not complete the other questions.

Figure 1 illustrates the output rule. For clarity, it asks three questions about one rate difference, using different thresholds. The current evidence places that difference between 0.07 and 0.21: one threshold comparison is true, another false, and an equivalence question is unresolved. Both true and false can be valid answers; unresolved means that more evidence is needed. The request is complete only when all its questions are settled. Section II walks through the records, counts and arithmetic behind the figure.

An incomplete request also raises a diagnostic question. Did the sampling policy miss a useful allocation of reads, or would the chosen inference rules require more evidence than the budget allows? A record of one unsuccessful execution cannot answer this. CohortCert adds an *offline* analysis: after the full sampling paths are revealed, it constructs a completing allocation or checks a lower bound on the reads any such allocation needs. This separates allocation opportunities from cost barriers under the stated rules. The analysis uses future labels; the online policies never receive them.

Existing work provides important parts of the solution. Semantic query systems optimize expensive model operations [1]–[3]; Evergreen verifies aggregate and ordering

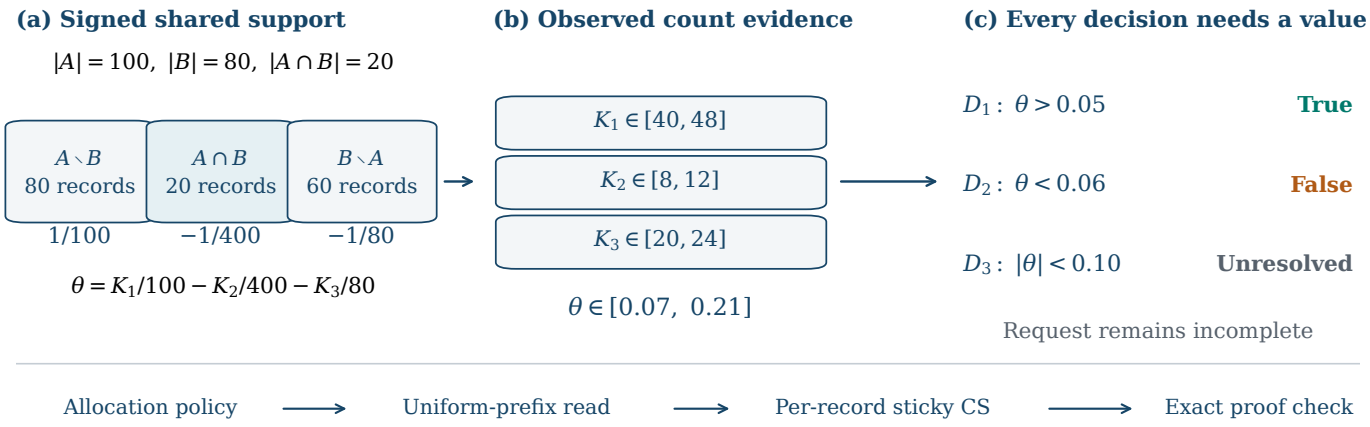


Fig. 1. One request, three questions. The same observed count ranges support a rate difference between 0.07 and 0.21. They justify the first answer as true and the second as false, but leave the third unresolved. The request therefore needs more reads. The three record groups contain records only in A, in both A and B, and only in B. All values are illustrative.

claims [4]. Boolean evaluation studies which observations settle a logical decision [5]–[7], and sequential inference supplies intervals that remain valid during adaptive sampling [8]–[11]. CohortCert connects these concerns at the level of a complete comparative request.

CohortCert makes three contributions. *Shared evidence:* a signed-count representation reuses labels across overlapping groups without losing the dependence created by a shared record. *Request-level execution:* a completion program combines the evidence needed by all requested decisions, while read-selection policies use that structure to choose the next batch. *Checked cost diagnosis:* independently verified upper and lower bounds identify completion opportunities and budget barriers on fixed sampling paths. Matched experiments separate the effects of interval propagation and read allocation.

The paper follows the same order as execution: Section II explains how counts support answers; Section III presents adaptive sampling and then offline cost analysis; Sections IV–V describe the experiments and results. The separate Technical Companion supplies full derivations, exact scores and reproduction parameters. Code, input manifests and both documents are public.[1]

II. Shared Evidence and Supported Answers

A. Task and Cost Model

The input is a fixed collection of texts, their group memberships, a binary label to obtain, the questions to answer, a call budget B and an allowed sampling-error probability α. Examples of labels are "contains toxic language" and "expresses joy." Group membership is already known; labels on unread records are unknown. The label source is part of the input: crowd annotations and model outputs can give different rates on the same texts.

[1]https://github.com/yzd2002/CohortCert. Original evidence archives are fixed at commit `aaele7b`; current manifests identify the manuscript and Technical Companion.

For comparisons and equivalence questions, the output is true, false or unresolved. For a completed ranking, it is the selected set of groups. The output also records the supporting evidence and cost. A request is complete only when every item has a supported answer. Strict rankings require the selected groups to beat every excluded group; a tie at this boundary can prevent completion even after all labels are read.

In the main experiments, one charged *read* is a submitted attempt to obtain one record's label for one property. Pilot reads and failed attempts count. The label is reused across all comparisons in that request. A different property or request is charged separately. Known metadata, text-feature preparation and physical model runtime are accounted for separately. Section IV distinguishes these replay costs from vector-label reads and live model calls.

B. A Worked Example

Let A contain 100 comments and B contain 80, with 20 comments in both. There are three disjoint sets: 80 comments only in A, 20 in the intersection and 60 only in B. Let K_1, K_2, K_3 be their unknown positive-label counts. The rates are $(K_1 + K_2)/100$ and $(K_2 + K_3)/80$. Subtracting gives

$$\theta = \mu_A - \mu_B = K_1/100 - K_2/400 - K_3/80. \qquad (1)$$

Here μ_A and μ_B denote the two rates. The negative overlap coefficient has a simple meaning: a positive shared comment raises both rates, but raises the smaller group's rate more. Its net effect is $1/100 - 1/80 = -1/400$.

Suppose sampling has supplied count ranges $[40, 48]$, $[8, 12]$ and $[20, 24]$. The smallest possible difference uses the smallest positive contribution and the largest negative contributions: $40/100 - 12/400 - 24/80 = 0.07$. The largest is $48/100 - 8/400 - 20/80 = 0.21$. Thus the evidence supports $\theta \in [0.07, 0.21]$.

This answers the three illustrative questions in Fig. 1. The entire range is above 0.05, so "more than five percentage points higher" is true. The range is above 0.06, so "less than six

TABLE I
KEEP THREE LEVELS SEPARATE: RECORD COUNTS, ANSWERS SUPPORTED BY THOSE COUNTS, AND THE NUMBER OF READS NEEDED TO SUPPORT ALL ANSWERS.

Symbol	Meaning
n_g, μ_g	Size and positive-label rate of cohort g.
N_h, K_h	Size and unknown positive count of residual cell h.
n_h, s_h	Records read and positives observed in that cell.
$[\ell_h, u_h]$	Current range for its total count K_h.
$Z_j, [L_j, U_j]$	Score of comparison j and its possible range.
$\underline{C}, \overline{C}$	Lower and upper bounds on total completion cost.

points higher" is false. It crosses 0.10, so "within ten points" remains unresolved. Two valid answers still leave the three-item request incomplete.

If we instead bound the two rates separately, we obtain $[0.48, 0.60]$ and $[0.35, 0.45]$, whose difference is the wider range $[0.03, 0.25]$. That calculation allows the shared count to take incompatible endpoint values in the two rates. Combining its contribution before bounding avoids this loss. We call this *signed propagation*: each shared count enters with its net positive or negative coefficient. The example illustrates interval arithmetic, not measured read savings.

C. Shared Counts and Comparison Bounds

For a larger request, records are grouped when their labels have the same contribution to *every* requested comparison. We call each such disjoint record group a *cell*. One count per cell is enough, even when many cohorts and questions use it. Records with zero contribution everywhere are excluded from the effective support I_q, the set of records that can affect request q.

The executor first takes a charged uniform pilot sample. Observed pilot contributions are known constants. The remaining records form residual cells, and a predictor fitted once on the pilot helps plan reads. Its parameters stay fixed during the run; its predictions do not count as observed labels. The cohort denominators remain the original group sizes.

A threshold comparison can be written as an integer score greater than zero. For nonempty cohorts and rational thresholds, subtract the threshold, include the known pilot contribution, then multiply by a positive common denominator:

$$Z_j = \beta_j + \sum_h a_{jh} K_h > 0. \tag{2}$$

The constant β_j contains the threshold and pilot contribution; a_{jh} is cell h's signed weight. Scaling changes neither the question nor its boundary. For example, $\theta > 0.05$ in (1) becomes $4K_1 - K_2 - 5K_3 - 20 > 0$ before any pilot is removed.

The score's smallest and largest possible values follow exactly the worked example:

$$L_j = \beta_j + \sum_h \min\{a_{jh}\ell_h, a_{jh}u_h\},$$
$$U_j = \beta_j + \sum_h \max\{a_{jh}\ell_h, a_{jh}u_h\}. \tag{3}$$

Since the score is integer-valued, $L_j \geq 1$ proves the comparison true and $U_j \leq 0$ proves it false. Otherwise it is unresolved. Equality is false for a strict comparison.

D. Decision Certificates

One threshold comparison is called an *atom* in the formal program. Several atoms may be needed for one answer. For example, "the rates differ by less than τ" requires both $\theta + \tau > 0$ and $\tau - \theta > 0$. Proving both establishes equivalence; proving either false establishes nonequivalence.

A *witness* is a complete route to a true answer. Consider "either A or D beats both B and C." Its two routes are $(A > B, A > C)$ and $(D > B, D > C)$. One fully supported route proves the decision true. To prove it false, one comparison must be false in *each* route. An unresolved comparison proves neither outcome. For top-k, each candidate set has exactly k members, and its route consists of all comparisons between selected and excluded groups. Internal ordering is unnecessary; only a positive, set-identifying answer is allowed.

The comparisons actually used to support all outputs form the *decision certificate*. This name means a checkable explanation of the answers. The checker uses individual comparison ranges. It does not make every possible joint logical deduction: two individually unresolved comparisons can still be logically incompatible. All completion-cost bounds below refer to this stated checking rule.

E. Confidence Sequences and Failure Handling

The system chooses a cell to read next, but cannot choose a favorable unread label inside it. Each cell has an independently uniform shuffled order; a read reveals the next record without replacement. A *prefix* means the first n records of that order. After observing s positives, the cell's total count must lie between s and $s + N_h - n$.

We tighten this logical range using the established finite-population confidence-sequence method [11]. Unlike an interval valid only at a prespecified sample size, this range is designed to remain valid while the system repeatedly chooses where to sample and when to stop. Conditional on the fixed labels, pilot and cells, the guarantee is

$$\Pr\{\text{all cell-count ranges cover throughout the run}\} \geq 1 - \alpha. \tag{4}$$

If all ranges cover, every answer supported by them is correct for the chosen label source. Sharing one covered count across several comparisons needs no extra risk allocation. This is a per-request sampling guarantee; it does not measure model accuracy or promise joint coverage across an entire session of requests.

For reproducibility, the method tests each candidate total by a likelihood ratio: a frozen predictive label law divided by the candidate's exact without-replacement probability. For ten records with three positives, seeing labels $(1, 0)$ in that order has probability $(3/10)(7/9)$. The predictive law is a half-and-half mixture of Beta$(1, 1)$ and a pilot-fitted Beta prior with 20 added pseudo-observations, hence concentration 22. Cell risks

Algorithm 1 From an input request to supported outputs

Charge a uniform pilot; freeze cells, predictor, priors and risks.
Give each cell a uniform shuffled order and an initial count range.
loop
  if a persistent failure has occurred **then**
    Return error with all submitted charges.
  end if
  Use the ranges to check every requested answer.
  if every requested answer is supported **then**
    Return the complete answer and its evidence/cost.
  else if the budget or available records are exhausted **then**
    Return the unresolved items and cost.
  end if
  Select a cell using common initialization/exploration or the policy.
  Clip the batch to unread records and remaining budget; charge it.
  Read the next records; update count intervals after each label.
end loop

sum to at most α. TC1 gives the exact likelihoods, mixture and validity argument.

After every returned label, the new count interval is intersected with all earlier ones. Intervals therefore narrow. If the intersection becomes empty, the execution keeps a persistent error flag; later reads or a census cannot erase the failure. This rule is shared by all allocation policies.

III. Adaptive Sampling and Cost Analysis

This section separates two uses of the same evidence rules. Online sampling chooses the next records using labels already observed. Offline analysis uses complete sampling paths to diagnose how many reads could have sufficed. We first describe the online loop and policies, then the completion program and its checked cost bounds.

A. Shared Execution Loop

Every policy uses the same count intervals and answer checker. Only its choice of the next cell differs; Forecast may also shorten a batch to its retained plan. This separation lets us test acquisition without changing the evidence rules.

Completion is checked after a successful batch, although intervals update after every record. The full batch is charged even if an intermediate prefix suffices. A partial/failed batch or a persistent interval failure suppresses claims while retaining submitted charges; rendering failures retain charges too. Section IV gives the common batch and exploration settings.

B. Read-Selection Policies

Matched balances the number of post-pilot records read from each cell. *WidthGreedy* favors cells expected to narrow the remaining comparison ranges most per read. *Variance-Greedy* targets estimated variance reduction, adapting Neyman allocation [12]. All use the same relevant records. Uncertainty-based scores ignore settled decisions and refuted positive routes.

DifficultyGreedy asks which next batch is expected to bring the *whole request* closer to completion. A comparison is difficult when its interval is wide or its midpoint lies near the threshold to be proved. Specifically, its side-specific difficulty is $W/(2\max\{e, 10^{-4}\})$, where W is the interval width and e the midpoint's signed distance toward the desired side of the threshold. The opposite side uses $-e$.

Within one positive route, the policy combines unresolved difficulties using log-mean-exp, a smooth maximum that emphasizes the hardest comparisons, then chooses the easiest route. For a false answer it chooses the easiest false comparison per surviving route and combines those difficulties. A binary decision uses the easier true/false side; a ranking uses its positive side. Adding the remaining decisions' difficulties gives the request score $\mathcal{D}$. For each candidate batch, predicted width shrinkage keeps the midpoints and eligible routes fixed. The policy chooses the greatest decrease in $\mathcal{D}$ per charged read. Exact smoothing, normalization, shrinking rules and ties are in TC3.

For one comparison, the same distance-to-threshold factor applies to every candidate cell. DifficultyGreedy therefore ranks batches like WidthGreedy in exact arithmetic, including clipped batch sizes. Single-contrast equivalence has the same width ordering for equal-price candidates; floating ties can differ. Distinctive gains can occur when one read helps several decisions. This is the reason to evaluate multi-decision summaries separately.

C. Forecast-Based Scheduling

Forecast predicts how each cell's count interval might shrink, then searches for a cheap allocation that would finish the request *if those predictions were correct*. It starts from pilot-mixture posterior-mean totals, rounds half-up, clips to the current ranges and makes the predicted ranges nested, ending at predicted singletons. These hypothetical ranges never justify an output; only observed count evidence can do that.

A coordinate pass starts with the longest predicted prefixes and removes reads while preserving predicted completion. If even those longest prefixes do not complete the request, there is no feasible forecast plan. A cheaper mixed-integer-program (MIP) solution replaces the incumbent only after an exact feasibility check. A retained plan advances the cell with the largest unfulfilled-planned-reads/unread-records ratio, then unfulfilled reads, then smallest cell ID. A plan-selected batch cannot exceed its remaining planned reads. Common initialization/exploration takes priority; fallback retains a usable plan or uses DifficultyGreedy.

The planning grid is $\{0, 16, 32, 64, 128, N_h - n_h\}$, clipped and deduplicated. It uses one thread, a $50\,\text{ms}$ solver limit and $150\,\text{ms}$ soft limit, and replans after eight rounds or plan exhaustion. The period was selected from $1, 4, 8$ on

four earlier airline development queries. Soft deadlines may alter rerun actions. TC2–TC3 specify the full predictions and implementation.

D. Offline Completion Costs

The offline evaluator looks back after complete sampling paths are known. A path is the list of count intervals after zero reads, one read, two reads, and so on in a cell's fixed order. Choosing one stopping prefix per cell specifies an allocation of reads. The evaluator asks for the cheapest allocation that would support every answer under the same checker. It keeps only prefixes before the first persistent failure and rejects selected comparison assertions inconsistent with the full reference labels. We call the remaining proofs *source-valid*. Other count intervals need not all cover for a selected proof to be source-valid.

This distinction can diagnose an incomplete run. For a budget of 100 reads, a checked completing allocation at 95 proves that completion was possible on those paths. A verified lower bound of 105 rules out a within-budget proof under those rules. A bracket $[80, 110]$ settles neither question. Every reported total includes the charged pilot.

Write C^*_{src} for the least additional reads after the pilot. With pilot size $|P|$, our total bounds satisfy $\underline{C} \le |P| + C^*_{\mathrm{src}} \le \overline{C}$. Thus an upper bound within budget proves possibility; a lower bound above budget proves a barrier. A policy that completes with a valid proof reaches a feasible prefix allocation, while batching and exploration may add reads. Hindsight existence alone does not supply a causal policy capable of finding those prefixes without future labels.

E. Completion Program

The optimization follows three rules: choose one read option in each cell; require enough interval improvement for every comparison used; and combine those comparisons into an answer for each requested item. For intuition, suppose a route needs gains of three units and two units in its two comparisons. One cell's next read might supply both gains; otherwise two different cells may be needed. This is a schematic explanation of shared progress, not a measured cost saving.

To formalize the first rule, let option k for cell h read m_{hk} more records at cost $c_{hk} = m_{hk}$; include a zero-read option. Set $x_{hk} = 1$ when that option is chosen and 0 otherwise. The objective is

$$\min \sum_{h,k} c_{hk} x_{hk}, \qquad \sum_{k} x_{hk} = 1 \quad (\forall h). \tag{5}$$

All options refer to one fixed set of nested interval continuations. These can be observed paths for offline diagnosis or predicted paths for Forecast.

For the evidence rule, $g^+_{jh}(m)$ is the increase in comparison j's lower score endpoint after m reads in cell h; $g^-_{jh}(m)$ is the decrease in its upper endpoint. They follow the min/max calculation in (3). Nested ranges make the gains nonnegative and additive across cells. From the current $[L_j(0), U_j(0)]$, true/false proofs need gains $d^+_j = \max\{1 - L_j(0), 0\}$ and $d^-_j = \max\{U_j(0), 0\}$. Binary T_j, F_j select a true or false assertion; both can be zero for unused comparisons:

$$\begin{aligned} &\sum_{h,k} g^+_{jh}(m_{hk}) x_{hk} \ge d^+_j T_j, \\ &\sum_{h,k} g^-_{jh}(m_{hk}) x_{hk} \ge d^-_j F_j, \qquad T_j + F_j \le 1. \end{aligned} \tag{6}$$

For example, setting $T_j = 1$ requires sufficient gain to raise the lower endpoint to at least one.

For the decision rule, let J_w list the comparisons in route w, and $\mathcal{W}_d$ list the routes for decision d. Binary z_w selects a positive route; p_d, r_d select true/false answers:

$$\begin{aligned} z_w &\le T_j && (j \in J_w), \\ \sum_{w \in \mathcal{W}_d} z_w &\ge p_d, \\ \sum_{j \in J_w} F_j &\ge r_d && (w \in \mathcal{W}_d), \\ p_d + r_d &= 1. \end{aligned} \tag{7}$$

Read the constraints from top to bottom. The first two lines require one wholly true route for a true decision. The third requires a false comparison in every route for a false decision. The last requires a supported value for each item. For ranking, fix $r_d = 0$ to require set identification.

Proposition 1. On fixed nonfailed nested paths, these rules are feasible exactly when the selected prefixes complete the request under the stated comparison-by-comparison checker.

Proof. A completed request supplies the true assertions and routes, or the false assertions blocking every route, required by the program. Its additive gains satisfy the evidence constraints. Conversely, those constraints support every selected assertion, and the decision constraints assemble an answer to every item. □

The program is compact in its listed cells, options, comparisons and routes; it avoids enumerating all negative proofs. General optimization is still NP-hard. Primary nonranking decisions have one positive route and rankings allow only positive completion, so their measurements do not demonstrate the generic representation advantage for many-route negative decisions. TC2 gives sizes, the Set Cover reduction and full gain formulas.

F. Verified Bounds and Overlap Analysis

A feasible allocation on actual prefixes gives an upper bound. For a lower relaxation, an optimistic grid option charges the earlier grid endpoint's cost but supplies the later endpoint's gain. Moving that option to the later actual prefix makes its gain available, at an added cost no greater than the grid width. This moves *prefixes*, rather than rounding an already integer cost. Linear-programming (LP) gaps can also contain integrality and solver gaps.

The numerical solver's reported bound is not accepted on trust. We convert its candidate dual multipliers to nonnegative rational numbers and recompute a weak-duality lower bound exactly, including the one-option-per-cell constraints. Feasible

upper allocations are checked separately with integer arithmetic. TC2 provides the full expression and the distinction from the unrestricted frontier that omits source-truth filtering. Wide or unavailable bounds remain explicit.

Finally, the overlap experiment changes only how count intervals are combined. The physical reads, sampling paths, pilot and support filtering stay fixed, and a shared record is still charged once. Propagating its two appearances separately widens the running example by 0.04 on each side, from $[0.07, 0.21]$ to $[0.03, 0.25]$. If a signed upper cost is below the relaxed lower cost, the separate propagation provably needs more evidence. This intervention isolates interval propagation; it does not compare shared acquisition with two independently sampled systems.

IV. Experimental Setup

The experiments separate three questions: which read-selection policy finishes more requests, what offline cost bounds explain about unfinished requests, and how reads translate into real model work. Table II distinguishes the studies. A *saved-label replay* supplies a policy with the stored label whenever it requests a record. It allows policies to follow the same sampling orders without paying for new model inference on every repetition. Only the live study measures fresh physical model calls.

A. CivilComments Records and Labels

CivilComments contains online comments with existing annotations about toxic language and content mentions. We use the identity-enriched test split distributed with WILDS [13]. A salted metadata hash selects $30{,}000$ rows eligible for at least one of eight mention-based cohorts (mention score ≥ 0.5). Selection does not use the label being compared or any method's outcome. Duplicate texts retain distinct source IDs.

These groups describe *what the comments mention*, not the identities of their authors or a representative demographic population. A comment may mention several topics and belong to several groups. Sizes range from $2{,}011$ to $9{,}409$ records, and all 28 pairwise intersections are nonempty (98–$3{,}291$ records). Of the selected rows, $8{,}767$ (29.22%) have multiple memberships (Fig. 2). This supplies natural overlap for testing shared counts.

We compare four binary properties: toxicity, insult, identity attack and obscene language. In the *crowd reference*, an existing annotation score at least 0.5 is positive, including ties. In the *model reference*, pinned Qwen3-8B [14] labels the same texts for the same four properties. They are two label targets on one population, not two independent datasets.

The model operator fixes prompts, weights/tokenizer, BF16, SDPA and a nonthinking template. Each record uses one forward pass and next-token $0/1$ scoring, with ties positive; generation and KV/result reuse are disabled. All contexts fit $4{,}096$ tokens without truncation. Small canary checks test operation and stability separately. The primary model experiments replay the resulting $30{,}000 \times 4$ saved label matrix. Agreement with crowd labels is evaluated separately from sampling validity.

B. Request Families

Table III explains the request families. An equivalence answer uses a tolerance of five percentage points. Dominance and mixed summaries contain three decisions, so answering just one does not count as success. Rankings ask for the best one or two of four groups; they are secondary endpoints.

Across the four properties there are 464 nonranking requests and 64 rankings, or 528 in total. Ten sampling seeds give the main denominator of $4{,}640$ per method/reference and $1{,}280$ for three-decision summaries. No request is removed because of its truth value or apparent ease of certification.

C. GoEmotions Records and Labels

GoEmotions consists of Reddit comments annotated for emotions [15]. Its three checksum-verified release files contain $58{,}011$ distinct IDs; $57{,}877$ have at least two clear, distinct raters. A metadata hash selects $30{,}000$ eligible rows without using emotion scores or outcomes. Anger, joy, sadness and fear are positive when at least half of the fixed clear raters select them, including ties. This study uses crowd labels only.

The eight cohorts use known metadata: the four most frequent subreddits among the selected rows, length/time at least their selected-row medians, and question/exclamation marks. Exact names, cutoffs and cohort sizes are in TC3 and the public metadata. The same 132 cohort-index templates transfer to each emotion with ten fresh seeds. This tests another text source and label task while retaining the request design.

D. Shared Execution Settings

Every policy has cap $B = \lceil 0.5|I_q| \rceil$ and request risk $\alpha = 0.05$. Cell risks are equal, with downward rounding so their sum is at most α. The charged pilot size is $\min\{256, \lfloor |I_q|/10 \rfloor, \lfloor B/4 \rfloor\}$. A pilot-only logistic model uses label-free TF–IDF text features compressed to 128 dimensions by SVD. Empty/single-class or failed fits use a smoothed pilot probability. TC3 gives the frozen prior shapes and all fitting parameters.

Initialization takes eight records on the first visit to a cell; later batches use at most sixteen. Every fourth round explores a uniformly selected nonexhausted cell. Other initialization rounds choose the largest unvisited cell, ties by largest ID. Policy ties and fallback rules are fixed; DifficultyGreedy chooses smallest ID within its gain tolerance and, for nonpositive gain, largest cell population then smallest ID. Batches are clipped to unread records and budget. Only Forecast can additionally clip a plan-selected batch to the remaining plan.

Section III explains Matched, WidthGreedy, VarianceGreedy, DifficultyGreedy and Forecast. *MaxDifficulty* replaces summed progress across decisions by the maximum on the 128 summary requests; it retains shared labels and inference. *Boundary*, used on CivilComments rankings, selects a tentative top set by relative estimated means and targets the weakest inside lower bound against the strongest outside upper bound. TC3 fixes its origin, ties and fallback. It is a scoped adaptation, not an implementation of every canonical ranking algorithm.

TABLE II

STUDY MAP. ONE EXECUTION IS ONE REQUEST, SEED, LABEL REFERENCE AND POLICY. NUMBERS FROM DIFFERENT ROWS HAVE DIFFERENT MEANINGS AND COST UNITS. THE MATHEMATICAL CONSTRUCTION IN SECTION VII IS SEPARATE FROM THESE MEASUREMENTS.

Study	Question	Scope	Unit of acquisition or measurement
Allocation replay	Which policy completes more requests?	73,760 executions	One record's label for one property.
Offline bounds	Could a proof fit the budget on these paths?	192 selected cases	Total reads including the pilot.
Earlier-corpus replay	Do patterns recur on earlier saved labels?	9,600 executions	One record–property label attempt.
MAST programs	What changes with the inference architecture?	7 programs, 8 paths each	One record revealing 14 flags at once.
Live inference	What are actual model and time costs?	23 query–policy executions	Forwards, formatted tokens and wall time.

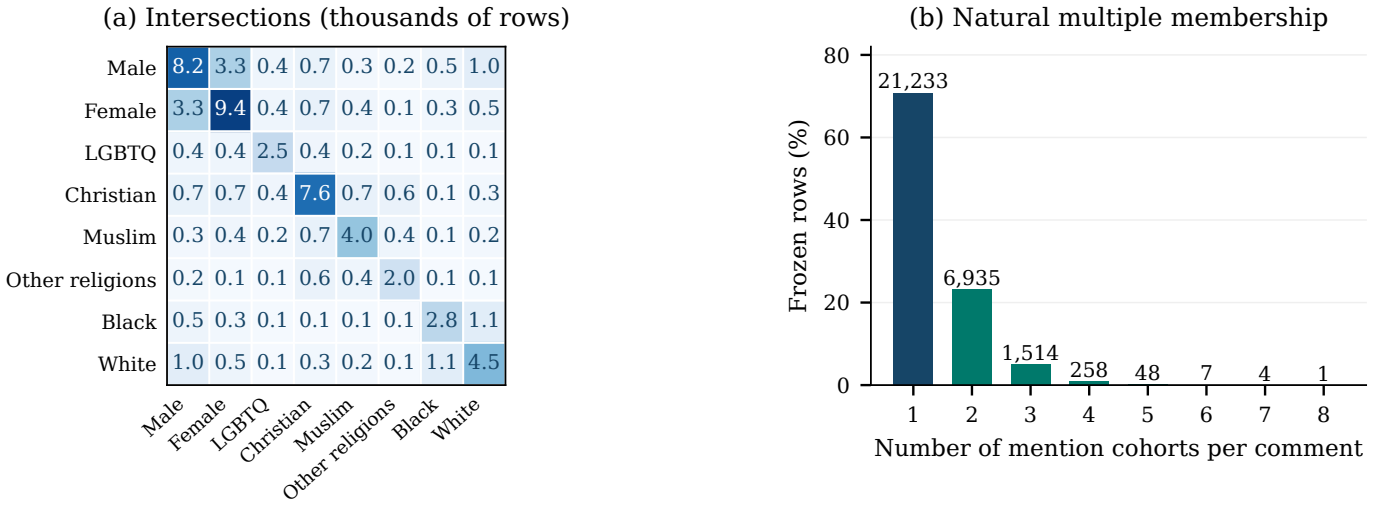


Fig. 2. Where sharing comes from in CivilComments. The heatmap shows cohort sizes on the diagonal and shared records off the diagonal, rounded to thousands. The bar chart counts each of the 30,000 records once by its number of memberships. These are metadata summaries; target labels are not used to select the records or form this figure. Exact counts accompany the artifacts.

TABLE III

REQUESTS PER PROPERTY. THE FIRST FOUR ROWS GIVE 116 PRIMARY REQUESTS; THE LAST GIVES 16 SECONDARY RANKINGS. FOUR PROPERTIES AND TEN SEEDS YIELD 4,640 PRIMARY AND 640 RANKING EXECUTIONS PER POLICY/REFERENCE.

Family	Count	What one request asks
Comparison	56	Is one cohort's rate higher?
Equivalence	28	Are two rates within 0.05?
Dominance summary	16	Does an anchor beat three other cohorts?
Mixed summary	16	One comparison and two equivalences over six cohorts.
Ranking	16	Top-1 (7 requests) or top-2 (9) among four cohorts.

The 73,760 replays comprise 31,680 original CivilComments executions, 1,280 Boundary executions, 23,680 stronger-control executions and 17,120 GoEmotions executions. Original inputs, queries, operator and three policies were internally hash-frozen before target conversion/inference. Stronger controls were added after those outcomes but code-frozen before their runs. GoEmotions was frozen before target conversion after CivilComments was known. These are internal freezes, not external preregistration.

E. Evaluation Metrics

A successful execution must return every requested answer and use only comparison assertions that agree with the full fixed label reference. A correct answer supported by a wrong selected assertion does not qualify. An interval excluding a true cell count is a separate event: it need not make the final answer wrong. Unresolved outputs, persistent failures and execution errors remain in every planned denominator.

We report submitted labels divided by effective-support size $|I_q|$, averaged over *all* executions, including early failures. A value 0.25 means one label attempt per four relevant records, approximately half the declared 50% cap. This quantity differs from a fraction of the cap and from physical runtime. A sensitivity charges errors/invalid proofs to the cap. Costs among jointly completed pairs are secondary and state their denominator.

Methods share pilot, cell-order, exploration and planning streams. Ten whole-workload seeds supply independent blocks; shared records, reversed comparisons and templates

make executions within a block dependent. Paired 95% percentile intervals resample these ten blocks 20,000 times, conditional on the fixed source/workload. Balanced summaries weight four properties and three families equally (pairwise, equivalence and three-decision summaries). MAST's eight paths are reported descriptively. An analyzer independent of the executor/compiler reconstructs rates, checks answers, selected assertions and submitted costs, and scans available count paths.

F. Diagnostic and Runtime Studies

Offline bounds. Salted hashes and support strata select 48 nonranking IDs across four properties, four subtypes and three strata without using outcomes. Two seeds and two CivilComments label references give 192 cases paired to the original three policies. Sampling paths, risk and source-validity restrictions match. Unknown or wide bounds remain. Relative gap is $(\overline{C}-\underline{C})/\max\{1,\overline{C}\}$ on total costs. The overlap intervention keeps the same reads and paths. TC3 defines cap-charged cost headroom and solver limits; no online acquisition is rerun.

MAST vector labels. MAST-Data [16] provides traces of multi-agent systems with released binary judge flags. We take 100 traces each from MetaGPT/GPT4o (A), MetaGPT/Claude (B) and ChatDev/GPT4o (C). Reading one trace reveals all 14 flags, so this cost unit differs from the scalar-label studies. The fixed request asks whether A's rate exceeds B's and C's for every flag: 28 comparisons, with ties false. Released judge labels are the reference, not human gold.

Seven configured programs each have eight paths; budgets $75, 150, 225, 300$ are four readouts of each path. Local RR/Width/DWidth use 42 count sequences with five charged pilots per cohort and risk $1/840$ per stream. NativeExact uses Evergreen's seeded exact aggregates [4]; NativeProgressive intersects its synchronized full-population intervals over time. Joint RR/LogGreedy instead use predictable without-replacement Kelly bets and verified composite-null witnesses [9], at risk $1/560$ per comparison shared by its two signs. TC3 defines these different inference rules and prefixes.

Credit requires verified normal closure, known within-budget cost, 28 correct supported outputs, and no earlier true-count exclusion or erased error. Interrupted/unverifiable paths remain uncredited with unknown cost tails; all four censors, including two Native startup censors, remain without retries. This history requirement is stricter than the main endpoint. Vector reads are not model forwards or tokens.

Earlier corpora and live inference. Earlier saved-label replications use 40,000 CFPB consumer narratives [17], [18] and 18,569 OpinRank reviews [19], four properties each. The combined 480 requests split 80/80/320 into train/development/test; 320 test requests, three policies and ten seeds give 9,600 revised replays at 50% caps. These previously examined labels have an unpinned upstream model and no human validation of the eight properties.

The live schedule uses five CivilComments queries and one 14,640-row airline ranking [20], selected by metadata hashes before method-outcome inspection, after primary crowd results were known. Six queries times three policies plus Boundary on two rankings give 20 warm executions. Three fresh-process executions repeat one CivilComments toxicity comparison, once per primary policy. Forwards, tokens, query time and lifecycle time are recorded separately. README and the Technical Companion document all external-input requirements and reproduction scope.

TABLE IV
ALL PRIMARY NONRANKING UNITS: 4,640 PER ROW. CIVIL C/M: CIVILCOMMENTS CROWD/MODEL; GO C: GOEMOTIONS CROWD. CORRECT REQUIRES EVERY VALUE AND SELECTED PROOF TO MATCH ITS REFERENCE. CALLS/$|I_q|$ AVERAGES ALL UNITS. E/W COUNTS EXECUTION ERRORS/SOURCE-WRONG PROOFS, RETAINED AS UNSUCCESSFUL.

Reference	Allocator	Correct	Calls/$\lvert I_q\rvert$	E/W
Civil C	DifficultyGreedy	2,886	0.3265	8/0
Civil C	WidthGreedy	2,850	0.3283	9/0
Civil C	VarianceGreedy	2,735	0.3363	10/0
Civil C	Matched	2,548	0.3456	28/0
Civil C	Forecast	2,784	0.3365	10/0
Civil M	DifficultyGreedy	3,384	0.3125	24/1
Civil M	WidthGreedy	3,330	0.3144	24/1
Civil M	VarianceGreedy	3,133	0.3253	22/0
Civil M	Matched	2,892	0.3326	37/1
Civil M	Forecast	3,158	0.3242	20/1
Go C	DifficultyGreedy	2,896	0.3084	3/0
Go C	WidthGreedy	2,894	0.3090	3/0
Go C	VarianceGreedy	2,772	0.3178	5/0

V. RESULTS

We first examine online completion, then separate the two sharing mechanisms and use offline bounds to explain incomplete requests. The remaining results examine prediction quality, label meaning and physical model work. These studies use the distinct units defined in Table II.

A. Whole-Request Completion

Table IV counts success only when the entire request finishes with correct reference-supported answers. For each row, all 4,640 planned executions remain, including unresolved and failed ones. The cost column is labels divided by relevant-record count, not elapsed time or the fraction of the budget used.

On CivilComments, DifficultyGreedy completes 36 more crowd-reference and 54 more model-reference requests than WidthGreedy. These are gains of 0.78 percentage points (95% block interval $[0.69, 0.86]$) and 1.16 points ($[0.91, 1.44]$). Mean all-execution read costs fall by 0.54% and 0.60%. Against VarianceGreedy, completion gains are 3.25 and 5.41 points and read reductions 2.89% and 3.94%; against Matched, read reductions are 5.51% and 6.04%. Shared labels and inference hold the evidence rules fixed.

The extra completions arise in multi-decision summaries. DifficultyGreedy and WidthGreedy agree on calls, status, answers/proofs and errors in all 6,720 paired CivilComments

comparison/equivalence units. Observable agreement need not mean identical read order. It is consistent with the single-comparison reduction in Section III.

GoEmotions shows a near tie: DifficultyGreedy completes 2,896 requests and WidthGreedy 2,894 of 4,640. The gain is 0.043 points ($[-0.065, 0.129]$), with 0.20% fewer reads. Against VarianceGreedy, its gain is 2.67 points with 2.98% fewer reads. All 3,360 paired comparison/equivalence units agree on measured outcomes. The audit retains 14 persistent interval errors across 17,120 executions and no source-wrong proof.

Forecast improves over Matched overall, but trails DifficultyGreedy by 2.20 crowd and 4.87 model completion points on CivilComments and uses more reads. Its crowd summaries improve markedly over Matched, whereas comparison and equivalence counts decline slightly. Equal weighting of properties and request families preserves the main ordering. Detailed per-family counts and secondary ranking results are retained in TC5.

B. Two Forms of Evidence Sharing

Progress across several decisions. On 1,280 CivilComments summaries per reference, summing DifficultyGreedy's gains completes 746 crowd and 907 model units. Replacing the sum by a maximum completes 728 and 894. The respective gains are 1.41 points ($[0.94, 1.95]$) and 1.02 points ($[0.39, 1.72]$), with 0.84% and 1.29% fewer normalized reads. On GoEmotions, the counts are 837 versus 833 of 1,280: 0.31 points ($[-0.08, 0.70]$) and 0.47% fewer reads. Thus the evidence for this aggregation benefit is clearer on CivilComments. Both variants already share labels and inference.

Combining an overlap before bounding. In 13 of 96 crowd and 17 of 96 model cases, the signed formulation's checked upper cost is below the separate-propagation formulation's lower cost. These 30 cases prove a necessary evidence-cost increase when overlap is propagated separately. Median lower bounds on extra reads in these subsets are 91 and 42. All 192 cases reconcile with the same physical prefixes and single charges. No case proves a crossing of the 50% budget boundary, and no online acquisition is rerun for this intervention.

There are 191 finite relaxed brackets and one no-source-proof case. Their relative gaps have medians 7.62% for crowd and 7.16% for model labels, with maxima 54.40% and 61.37%. Uncertain bounds limit how often a strict cost difference can be established.

C. Offline Completion Opportunities

Table V separates possible completions, proved budget barriers and unresolved cost bounds. It uses the offline analysis, which knows the future sampling paths. Across 192 selected cases, it constructs within-budget source-valid proofs in 157, rules out such proofs within budget in 27, and finds one case with no source-valid proof on any legal prefix. Seven cases remain undecided. None is replaced or removed.

TABLE V
OFFLINE DIAGNOSIS OF ALL 192 SELECTED CASES. THE FIRST FOUR ROWS PARTITION THE 96 CASES PER REFERENCE. THE ONLINE POLICY COUNTS BELOW THE RULE ARE SEPARATE OBSERVATIONS. EVERY COST INCLUDES THE PILOT.

Diagnostic or policy	Crowd	Model
Within-budget proof constructed	67	90
Lower bound exceeds budget	24	3
No source-valid proof on any legal prefix	0	1
Bounds do not settle the budget	5	2
DifficultyGreedy completes	62	82
Forecast completes	61	75
Matched completes	52	66

DifficultyGreedy leaves 13 cases unfinished despite a constructed within-budget proof; Forecast leaves 21. These are opportunities revealed with hindsight, not guaranteed savings available to an online policy. Other noncompletions, especially for crowd labels, face proved cost barriers. Figure 5 keeps the uncertain cases visible.

Of 191 finite signed brackets, 103 reach a 5% relative total-cost gap; the remaining 88 reach gaps up to 63.14%. A precision pass retains rational dual multipliers without changing cases or upper proofs. Relative to DifficultyGreedy's cap-charged summed reads, bounded hindsight headroom is 7.03–12.59% on crowd and 13.55–23.97% on model labels. TC5 retains the separate forecast-optimism diagnostic and its source-coverage denominators.

D. Forecast Quality and Planning Cost

A cheaper predicted plan can still guide acquisition poorly. At 78,937 crowd planning states, MIP improves the forecast-feasible incumbent, yet Forecast does not beat the simpler DifficultyGreedy policy. This does not isolate MIP's causal effect. Mean saved-label replay CPU time is 1,272 ms for Forecast, 608 ms for DifficultyGreedy and 531 ms for Matched; these are planning/replay costs, not live-model time.

Earlier-corpus replications show the same overall policy ordering: among 1,600 units per method and corpus, Matched, DifficultyGreedy and Forecast complete 653, 721 and 701 CFPB requests, respectively. On OpinRank the corresponding counts are 228, 326 and 298. All 9,600 executions are audited, including an adverse shared path. TC5 gives all normalized costs, family breakdowns, rankings and the $9/94$ crowd and $6/91$ model forecast-optimism cases. These additions do not replace the primary endpoint.

E. Label References and Correctness

The statistical target is the chosen fixed label source. It is possible to have a correct answer with an unsupported intermediate assertion, or a supported model-relative answer that disagrees with crowd labels. We check these separately.

The original three-policy CivilComments grids retain 50 crowd and 100 model persistent interval errors. Three wrong model false-equivalence outputs come from one shared query/seed path. Additional controls retain their errors and

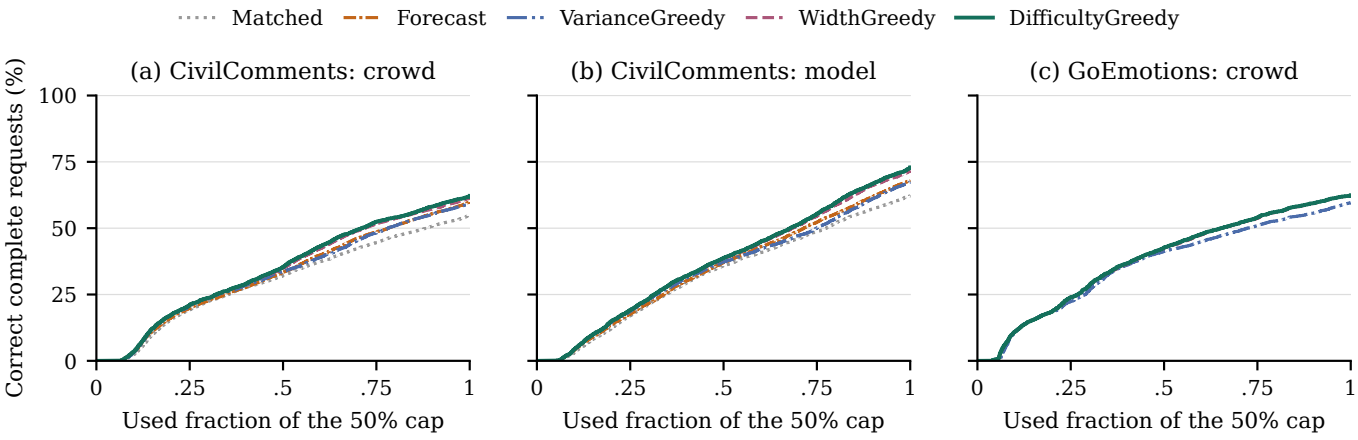


Fig. 3. How much of the budget is needed for complete answers? Each curve uses all 4,640 primary nonranking executions per policy/reference. The horizontal axis is the used fraction of the declared 50% cap; 1 is approximately 0.5 labels per relevant record. The vertical axis counts correct reference-supported completions. Unresolved, erroneous and wrong-proof runs remain uncompleted. These are stopping costs from fixed runs, not fresh runs at each budget. CivilComments uses two references on the same texts; GoEmotions is a second source.

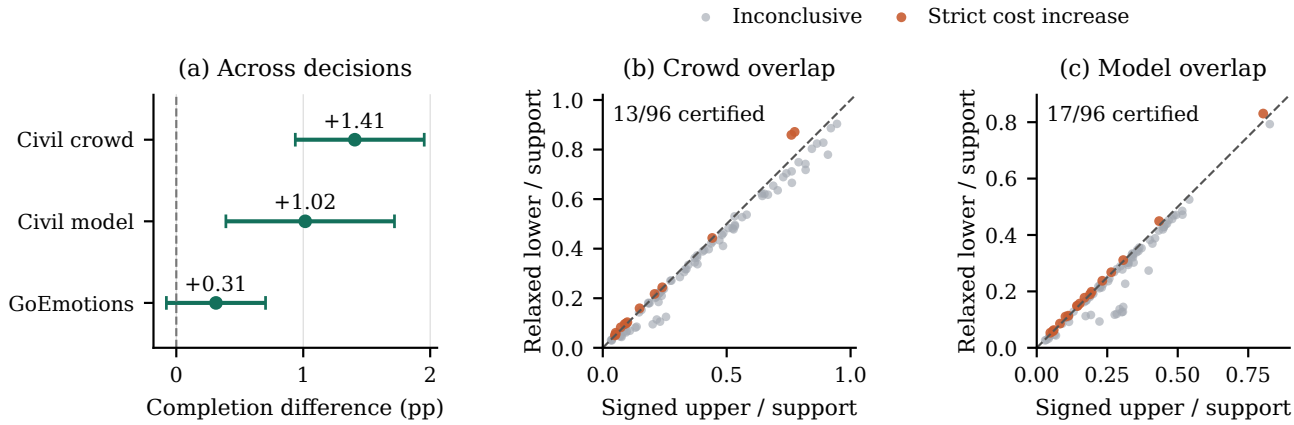


Fig. 4. Two separate sharing effects. Panel (a) compares summed versus maximum progress across decisions; bars are paired 95% seed-block intervals. Panels (b,c) compare the signed upper cost on the horizontal axis with the relaxed lower cost on the vertical axis, both divided by relevant-record count. A point above the diagonal proves that separate propagation needs more reads. Gray points are inconclusive. Only finite pairs have coordinates; the model case with no legal proof stays in the 96-case denominator and Table V. No case proves a budget crossing.

TABLE VI
PINNED MODEL LABELS RELATIVE TO EXISTING THRESHOLDED CROWD REFERENCES, ALL 30,000 ROWS PER PREDICATE. PERCENTAGES ARE DESCRIPTIVE; PRECISION/RECALL CONCERN THE POSITIVE CLASS.

Predicate	Agreement	Precision	Recall	Bal. acc.
Toxicity	83.03	48.16	36.79	64.48
Insult	74.86	19.65	70.81	73.01
Identity attack	74.39	17.25	69.46	72.11
Obscene	96.58	9.30	48.72	72.81

wrong proofs (Table IV); a shared CFPB path also has wrong values/proofs. Exact-rational checks allow these exclusions at nominal thresholds without identifying a numerical defect. These dependent paths are not an empirical calibration of the stated risk. No favorable retries erase them.

On 30,000 rows per property, crowd/model agreement is 74.39–96.58%, but positive precision is only 9.30–48.16%. For rare labels, high agreement can largely reflect agreeing on negatives. Operational canaries do not validate semantic fidelity. Prompts paraphrase source concepts rather than reproduce the historical questionnaires exactly; full confusion counts, recall, balanced accuracy and cohort breakdowns accompany the artifacts.

Even before sampling, the full references disagree on 108 of 464 nonranking answers and 13 of 64 ranking sets. Of DifficultyGreedy's 3,838 model-proof-valid completions across all 528 requests and ten seeds, 747 have a crowd-inconsistent answer or selected proof. Among 5,250 model-reference runs with a recorded terminal interval collection, 2,027 have a crowd contrast outside its model-target interval. These are reference changes, not additional noncoverage of the

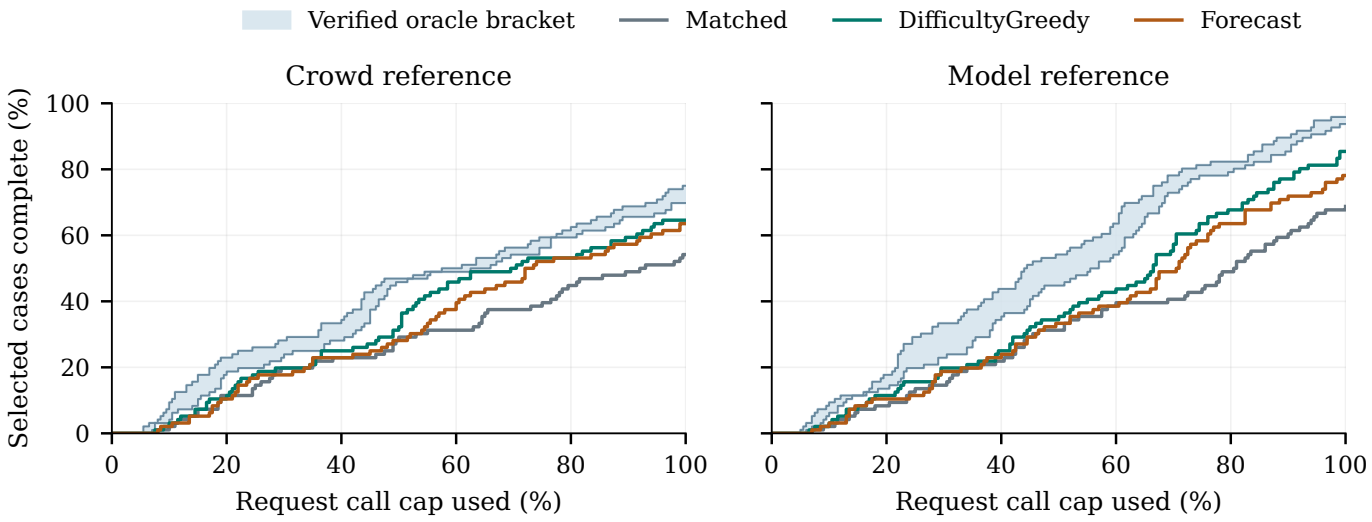


Fig. 5. What could fit within a given budget? Each panel retains 96 selected cases. The horizontal axis is percentage of the declared call cap. The shaded band's lower edge counts constructed proofs within that budget; the upper edge also includes cases not yet excluded by their lower cost bounds. This is deterministic cost-bound uncertainty, not a statistical confidence interval. Policy lines show complete-answer stopping costs from fixed executions. The offline evaluator knows future paths under the fixed inference rules.

TABLE VII
THE FIXED MAST REQUEST. COMPLETE: CREDITED PATHS OUT OF ALL EIGHT AT BUDGET 300. COST RANGES APPLY ONLY TO CREDITED PATHS. INTERRUPTED/UNVERIFIABLE PATHS RETAIN UNKNOWN COST TAILS; EARLIER COUNT EXCLUSIONS PREVENT CREDIT EVEN IF FINAL ANSWERS ARE RIGHT.

Program	Complete	Cost	Other paths
Local-CS / RR	7/8	298	1 censored
Local-CS / Width	8/8	278–288	None
Local-CS / DWidth	8/8	278–288	None
NativeExact adapter	7/8	300	1 censored
NativeProgressive / RR	5/8	300	2 prior exclusions; 1 censored
Joint-Kelly / RR	8/8	298	None
Joint-Kelly / LogGreedy	7/8	278–290	1 censored

fixed model target. The eight earlier-corpus properties remain without human-reference validation.

F. Vector-Label Inference

The MAST request compares three collections of agent traces on 14 judge flags. It is complete only after all 28 comparisons have correct supported answers. One read reveals a whole flag vector, so its counts cannot be compared directly with scalar-model calls. The seven configured programs test local count intervals, native aggregate intervals and joint comparison evidence on this same request. RR cycles through cells; Width and DWidth choose cells by uncertainty. DWidth is distinct from the main DifficultyGreedy heuristic.

All programs have zero credited completions at budgets 75, 150 and 225. At 300, Width and DWidth have identical counts and costs on all eight paths, so difficulty weighting adds no measured benefit here. Joint-Kelly/LogGreedy's cost range on seven credited paths cannot establish unconditional savings over Joint-Kelly/RR's eight.

A post-hoc import-context correction reverified the unchanged trajectories. NativeProgressive earns $5/8$: two normally closed paths have correct final answers but earlier true-count exclusions. After 99 reads from 100 records with 19 positives seen, the total can still be 19 or 20. Binary64 clipping/count readout nevertheless drops the upper count from 20 to 19 in those histories. A separate arithmetic audit finds such interface losses in 94 of 5,049 proper-prefix states; directed enclosures remove these enumerated losses. This counts deterministic states, not an empirical error rate. No inference trajectory is rerun. TC4–TC5 retain the full closure and verification flags.

G. Live Model Costs

The 23 live executions use pinned Qwen3-8B on one RTX 5090 (32 GB). A forward is an actual model evaluation; tokens measure its formatted input. Warm executions use the running model, while fresh-process lifecycle time also includes startup and exit. Feature preparation is separate, and fresh processes may still benefit from filesystem caches.

The runs produce 69,799 verified corpus forwards and 13,302,613 tokens, plus 192 setup-canary forwards and 17,888 tokens. Independent checks verify proofs and 63,911 interval updates. Correct completion is $17/20$ warm and $3/3$ fresh-process. All three unresolved warm policies exhaust the same insult-equivalence query's 2,766-read cap; they remain in the means below. Boundary's two rankings are separate.

Across five jointly correct warm query pairs, summed wall-time ratios are 1.2410 for Matched/DifficultyGreedy and 1.0356 for Matched/Forecast. This selected-pair comparison is secondary. All warm query times sum to 1,703.034 s; enclosing lifecycles sum to 1,731.961 s, including setup, I/O and teardown. These are single measurements in three chronological windows.

TABLE VIII
LIVE MODEL COSTS. WARM COLUMNS GIVE TOTAL FORWARDS AND MEAN QUERY SECONDS OVER SIX QUERIES PER POLICY, INCLUDING UNRESOLVED ONES. FRESH-PROCESS QUERY/LIFECYCLE SECONDS USE ONE COMMON CIVILCOMMENTS QUERY; LIFECYCLE INCLUDES STARTUP AND EXIT.

Policy	Warm total forwards	Warm mean seconds	Fresh process query/lifecycle (s)
Matched	21,903	94.78	40.153/66.254
DifficultyGreedy	18,125	78.68	37.687/64.455
Forecast	21,039	91.90	40.300/68.708

One exactly checked count exclusion recurs in three warm and three fresh-process dependent paths, while their final proofs remain source-valid. In total, $20/23$ executions complete correctly. Six full-support materialized lookups need no new forwards, but the missing controlled cold-build ledger prevents a materialization break-even claim.

VI. RELATED WORK

Reducing expensive model work. LOTUS, ZenDB, DocETL and Bond optimize semantic model operations in data-processing workflows [1], [21]–[23]. ABAE, InQuest, Thalamus and SUPG study approximation when predicates are expensive [2], [3], [24], [25]. These works motivate treating labels as a query-execution cost. Our unit of success is a complete set of supported cohort decisions, with shared count evidence and explicit completion-cost bounds.

Knowing when enough evidence has been collected. Evergreen verifies aggregate and ordering claims [4]; SemViews separates planning with prior labels from fresh certification [26]. Boolean evaluation and decision diagrams study observations that resolve logical decisions [5], [7], and query-guided resolution uses Boolean provenance [6]. Verification-error models address when to verify again [27]. CohortCert fixes a specific count-interval checker and asks which reads can finish all its requested decisions.

Other inference and ranking rules. PEAK pools evidence across streams [8]; stratified and target-aware methods handle composite objectives [9], [10]. Joint confidence regions and constrained selection can exploit relationships beyond separate count intervals [28], [29]. IFocus, LUCB and best-arm/multiple-answer identification focus effort on unresolved orderings or selection boundaries [30]–[33]. These motivate alternative architectures and controls. Our cost frontier is conditional on the chosen paths, risks and comparison-by-comparison proof rules; it is not a lower bound for every possible inference method.

VII. DISCUSSION

A within-budget offline proof identifies an acquisition opportunity: the fixed evidence paths contain enough useful information under the chosen checker. A lower bound above budget instead identifies a barrier for that checker. Neither result supplies an online strategy by itself, and changing the inference rules can change the minimum cost.

A separate construction makes this concrete. Three disjoint 100-record cohorts have 14 flags each, with 15 charged pilot reads. The request asks whether each of two cohorts beats a common third cohort on every flag (28 comparisons). With all-zero hidden labels and global risk 5%, the specified 42-stream local family needs at least 205 vector reads; an OR-composite proof of the same request needs at most 197. A different request, asking for rate gaps greater than five percentage points, uses $\mathrm{Beta}(1,1)$ and needs at most 171, with risk still 5%. These are constructed proof costs, not MAST measurements. TC2 supplies the definitions and derivation.

Alternative inference also needs a valid sampling law. TC2 diagnoses one displayed transfer to without-replacement sampling [34]; it leaves the published with-replacement result and other joint approaches intact. The measured scope here is eight cohorts, shared request templates, one MAST request and 23 live executions. Fixed-label sampling guarantees do not establish semantic fidelity, robustness to drift or session-wide coverage. Verification-error models [27] and batching/materialization [23], [26], [35] address complementary deployment questions.

VIII. CONCLUSION

CohortCert asks which shared reads are enough to answer every question in a comparative request. It turns overlapping records into count evidence, checks the requested answers, and uses a completion program to plan or diagnose read costs. The experiments distinguish modest allocation gains from provable overlap effects and hindsight opportunities, while keeping the label reference and inference rules explicit.

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